Abuthahir B

Data Scientist | ML Engineer

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PROFILE

Recent graduate with a B.E in Computer Science Engineering, skilled in data analysis, machine learning, and dedicated to extracting valuable insights from data. Solid foundation in **Data Science, Machine Learning, Natural Language Processing, and Generative-AI**, gained through a certification from GUVI, IIT Madras Research Park. Eager to contribute to a data science team with innovative, real-world solutions.

SKILLS

Python | SQL | Pandas | NumPy | Machine Learning | Deep Learning | NLP | LLM | RAG | FAISS | HuggingFace | Power BI | Streamlit | Data Visualization | Data Analysis

PROJECTS

Telecom AI Copilot (RAG-based Chatbot)
Key Skills: RAG, FAISS, HuggingFace, LLM, Semantic Search
- Developed a telecom AI assistant using RAG architecture for document-grounded customer support response generation
- Implemented semantic retrieval using HuggingFace's **all-MiniLM-L6-v2** model to convert queries and documents into vector embeddings
- Integrated **FAISS** for efficient vector similarity search across telecom policy and pricing documents, enabling fast and accurate document-grounded responses
- Designed an evaluation pipeline to measure retrieval accuracy and response quality of the LLM-based system

IMDb Movie Recommendation System (NLP)
Key Skills: Python, Selenium, NLP, TF-IDF, Cosine Similarity
- Built a movie recommendation system by scraping and analyzing **IMDb 2024** movie data using Selenium
- Applied **TF-IDF vectorization and Cosine Similarity** to recommend movies based on storyline similarity
- Developed an interactive Streamlit app that returns the **top 5 similar movie recommendations** for any user-input storyline

Swiggy Restaurant Recommendation System
Key Skills: Python, Pandas, Machine Learning, Streamlit
- Built a personalized restaurant recommendation system using **One-Hot Encoding and K-Means/Cosine Similarity** based on city, cuisine, rating, and cost preferences
- Deployed an interactive **Streamlit dashboard** for real-time recommendations and user-friendly data visualization

NutriClass: Food Classification Using Nutritional Data
Key Skills: Python, Scikit-learn, Machine Learning
- Built a multi-class food classification system using **Logistic Regression, Random Forest, SVM, XGBoost, and KNN** to predict food categories from nutritional attributes
- Performed end-to-end preprocessing, feature engineering, and evaluation using **Accuracy, Precision, Recall, and F1-score**

Data-Driven Stock Analysis Dashboard
Key Skills: Python, Pandas, SQL, Power BI, Streamlit
- Developed a dashboard to analyze **Nifty 50** stock performance, covering volatility, cumulative returns, sector-wise trends, and stock correlations
- Automated data pipelines converting YAML-based stock datasets into structured CSV/SQL formats for real-time insights on top gainers/losers and market trends

Client Query Management System
Key Skills: Python, MySQL, Pandas, Streamlit
- Developed a Client Query Management System with secure login authentication, query tracking, and status management for real-time client support
- Built an interactive **Streamlit dashboard** for support teams to monitor, filter, and close queries efficiently

- Analyzed query trends and resolution performance using **SQL queries and datetime-based tracking**

EDUCATION

IIT-M Advanced Programming Professional & Master Data Science 07/2025 – 02/2026
GUVI, IIT Madras Research Park Chennai,Tamil Nadu

Bachelor in Computer Science Engineering 06/2021 – 05/2025
Annaporana Engineering College Salem,Tamil Nadu

Diploma in mechanical engineering 06/2018 – 04/2020
Mahendra Polytechnic College Namakal,Tamil Nadu

CERTIFICATES

Data Science Program – GUVI, IIT Madras Research Park
Modules: Python | SQL | Pandas | NumPy | EDA | Machine Learning | NLP | Deep Learning | Data Visualization